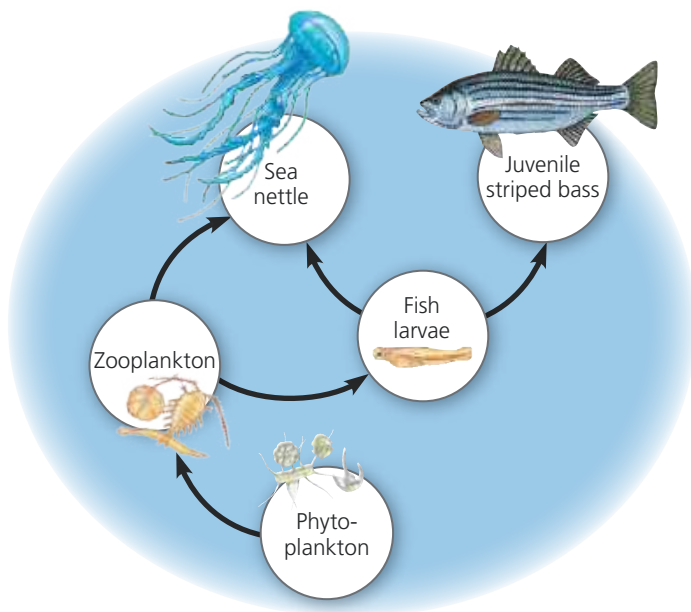


▼ **Figure 54.18 Partial food web for Chesapeake Bay estuary.**

The sea nettle (*Chrysaora quinquecirrha*) and the juvenile striped bass (*Morone saxatilis*) are the main predators of fish larvae.



VISUAL SKILLS Based on this food web, identify organisms that function as primary, secondary, or tertiary consumers.

Complicated food webs can be simplified in two ways for easier study. First, species with similar trophic relationships in a given community can be grouped into broad functional groups. In Figure 54.17, more than 100 phytoplankton species are grouped as the primary producers in the food web. A second way to simplify a food web is to isolate a portion of the web that interacts very little with the rest of the community.

Figure 54.18 illustrates a partial food web for sea nettles (a type of cnidarian) and juvenile striped bass in the Chesapeake Bay estuary on the Atlantic coast of the United States.

➔ **Mastering Biology** HHMI Video: Riverine Food Webs: How Flow Rates Affect Biomass



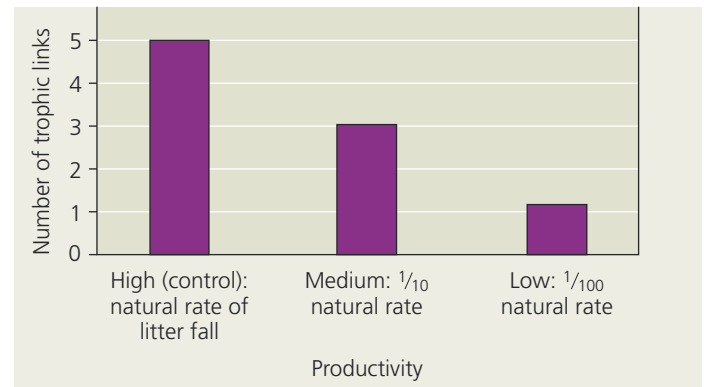
Limits on Food Chain Length

Each food chain within a food web is usually only a few links long. In the Antarctic web of Figure 54.17, there are rarely more than seven links from the producers to any top-level predator, and most chains in this web have fewer links. In fact, most food webs studied to date have chains consisting of five or fewer links.

Why do food chains tend to be relatively short? The most common explanation, known as the **energetic hypothesis**, suggests that the length of a food chain is limited by the inefficiency of energy transfer along the chain. On average, only about 10% of the energy stored in the organic matter of each trophic level is converted to organic matter at the next trophic level (see Concept 55.3). Thus, a producer level consisting of 100 kg of plant material can support about 10 kg of herbivore biomass and 1 kg of carnivore biomass. The energetic hypothesis predicts that food chains should be relatively

▼ **Figure 54.19 Test of the energetic hypothesis for the restriction of food chain length.**

Researchers manipulated the productivity of experimental tree-hole communities in Queensland, Australia, by providing leaf litter input at three levels. Reducing energy input reduced food chain length, a result consistent with the energetic hypothesis.



longer in habitats characterized by higher photosynthetic production, since the amount of energy stored in primary producers is greater than in habitats with lower photosynthetic production.

Ecologists tested the energetic hypothesis in experiments that mimicked the “tree-hole” communities found in tropical forests. Many trees have small branch scars that rot, forming holes in the tree trunk. These tree holes hold water and provide a habitat for tiny communities consisting of microorganisms and insects that feed on leaf litter, as well as predatory insects. **Figure 54.19** shows the results of experiments in which researchers manipulated productivity by varying the amount of leaf litter in an experiment using artificial tree holes (water-filled pots placed around the trees); previous studies had shown that the communities that colonized these pots were similar to those in natural tree holes. As predicted by the energetic hypothesis, the pots with the most leaf litter, and hence the greatest total food supply at the producer level, supported the longest food chains.

Another factor that may limit food chain length is that carnivores tend to be larger at higher trophic levels. The size of a carnivore puts an upper limit on the size of food it can take into its mouth. And except in a few cases, large carnivores cannot live on very small food items because they cannot obtain enough food in a given time to meet their metabolic needs. Among the exceptions are baleen whales, huge filter feeders with adaptations that enable them to consume enormous quantities of krill and other small organisms (see Figure 41.5).

Species with a Large Impact

Certain species have an especially large impact on the structure of entire communities because they are highly abundant or play a pivotal role in community dynamics. The impact of these species occurs through trophic interactions and their influence on the physical environment.